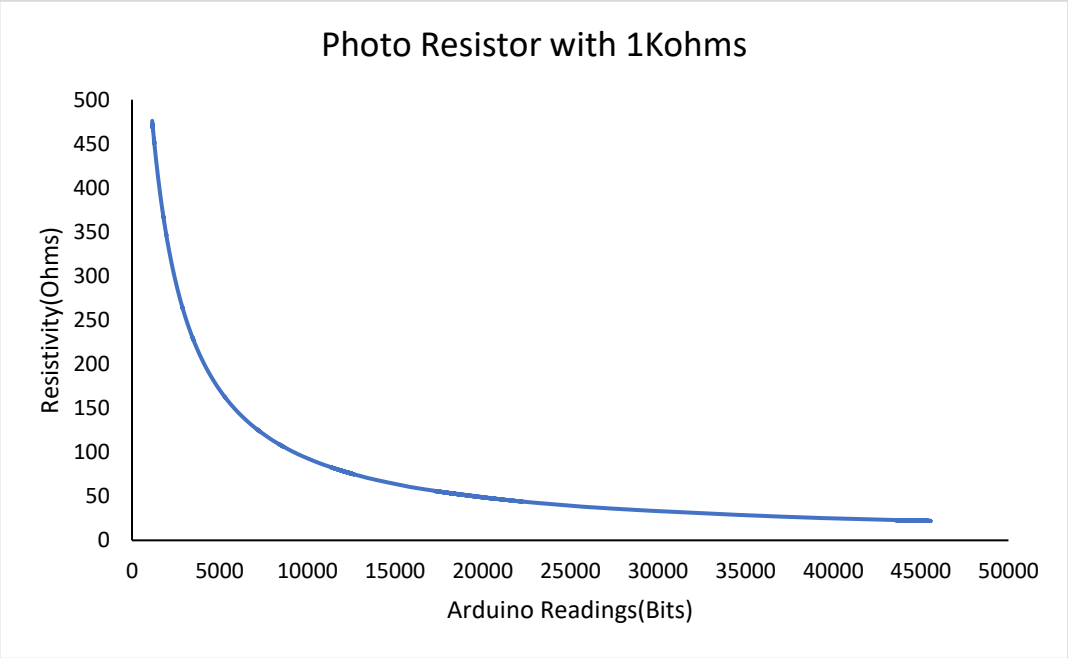
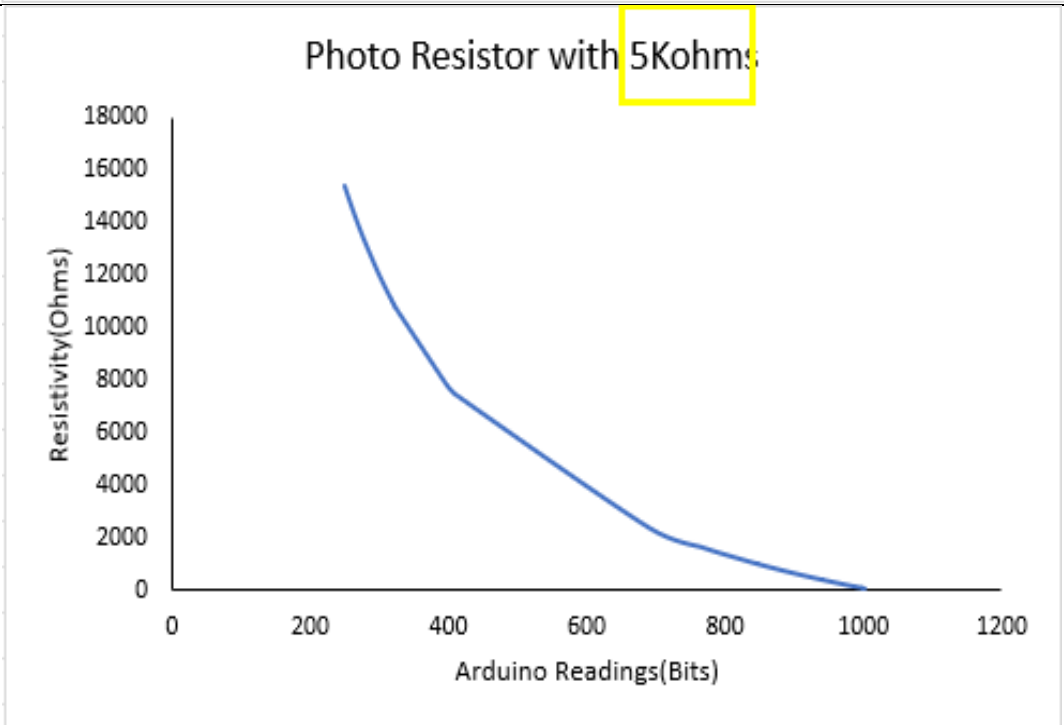
Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

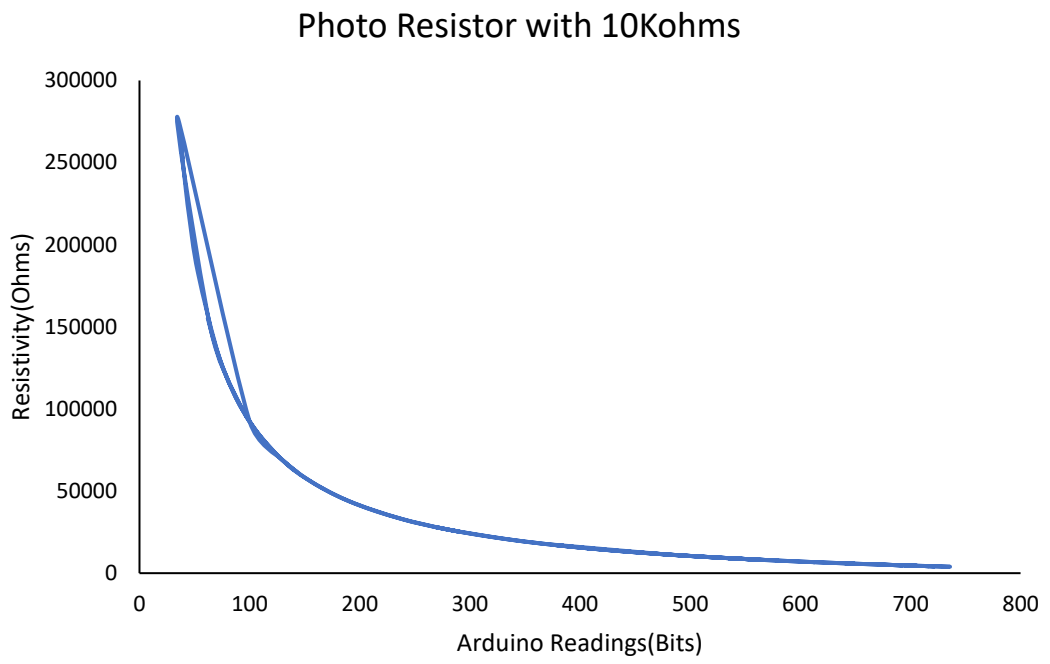
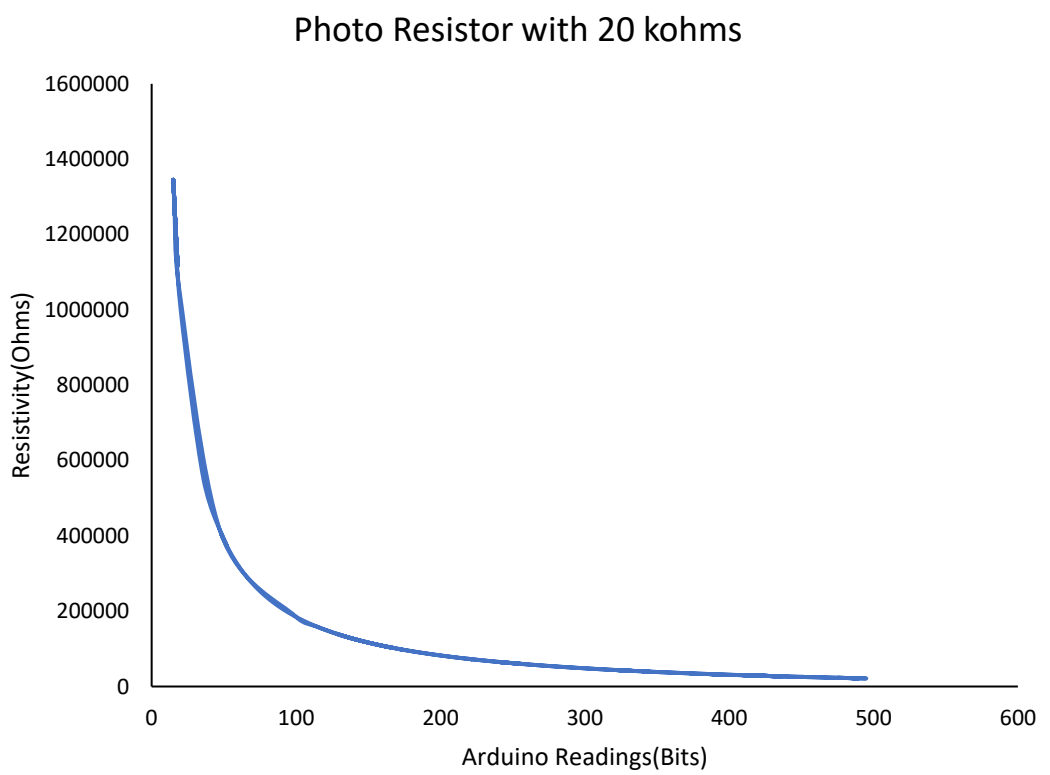
1. Resistor Selection for photo resistor. (34%)
    a. Create a graph for each of the estimated resistances of the photo resistor when pair with another passive resistor.
    b. Combine all the graphs and show the range of operation with a normalize voltage.
    c. Answer: Which passive resistor is better suited to be with the photo resistor? What is the range of the resistor? What makes the resistivity to go down?

1 kOhm Resistor	
5 kOhm Resistor	

Lab 4 Report

Name: Tyler William Grindrod

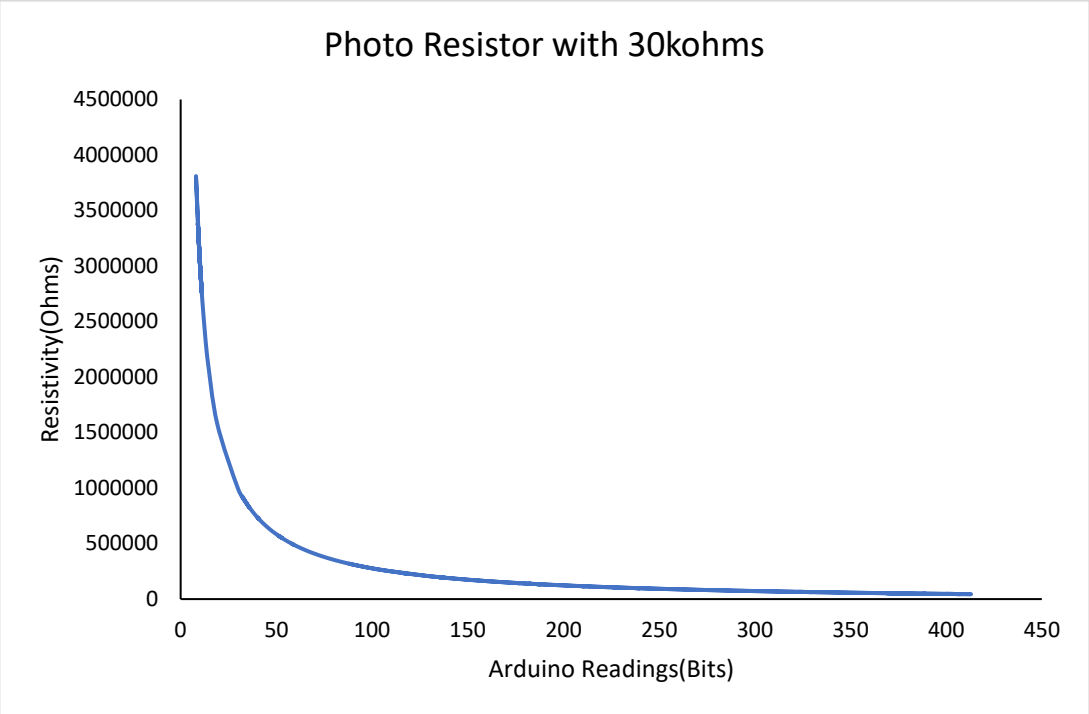
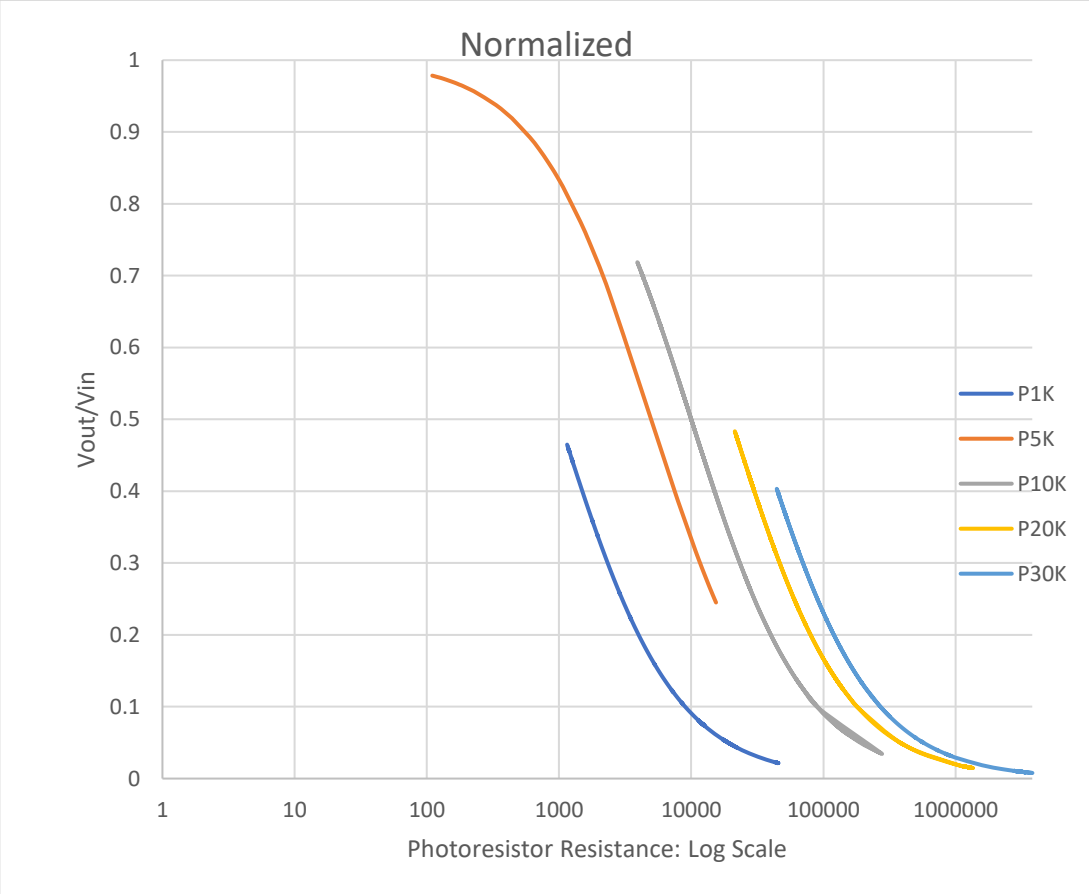
PID: 5695162 Date: 03/10/26

10 kOhm Resistor	
20 kOhm Resistor	

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

30 kOhm Resistor	
Normalized (combined)	

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

Answer Part C:	In Voltage Divider the maximum change in Vout occurs when the Resistor that is fixed closely matches the resistace of our sensor. When 1K/30K ohm fixed resistor is used, the (Vout/Vin) is close to 0(signal compressed to the bottom). From the normalized graph above, There is a larger response (higher sensitivity) to light from the **5k 0hm resistor**. So this our best suited fixed resistor. Its dynamic range is 751 ADC. The resistivity of the photoresistor goes down as the light intensity increases, allowing more current to flow and increasing the output voltage in this configuration.

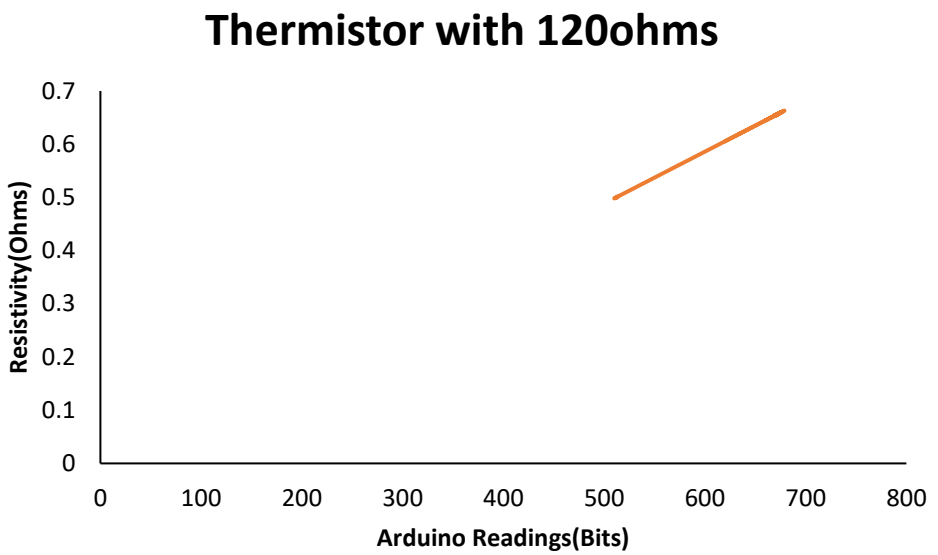
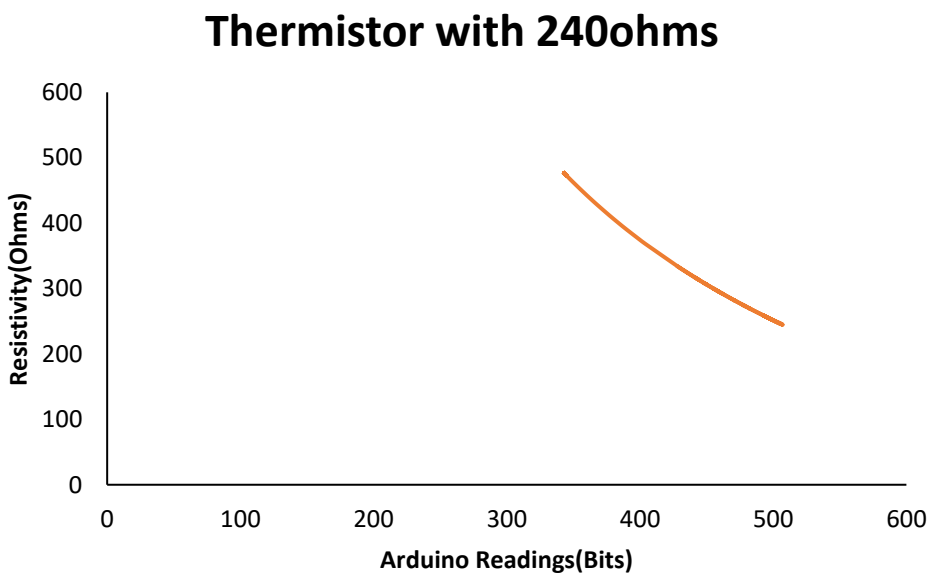
Code:	// Lab 5: PhotoResistor Sensor
	// Tyler W. Grindrod
	
	#define photoResistor_Pin A0
	
	void setup() {
	  Serial.begin(9600);
	}
	
	void loop() {
	  // Read PhotoResistor Analog-to-Digital Converter (ADC) values between 0-1023
	  // 0 = 0V, 1023 = 5V
	  int luminosity = analogRead(photoResistor_Pin);
	  Serial.println(luminosity);
	  delay(100); // wait 100ms between readings
	
	}

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

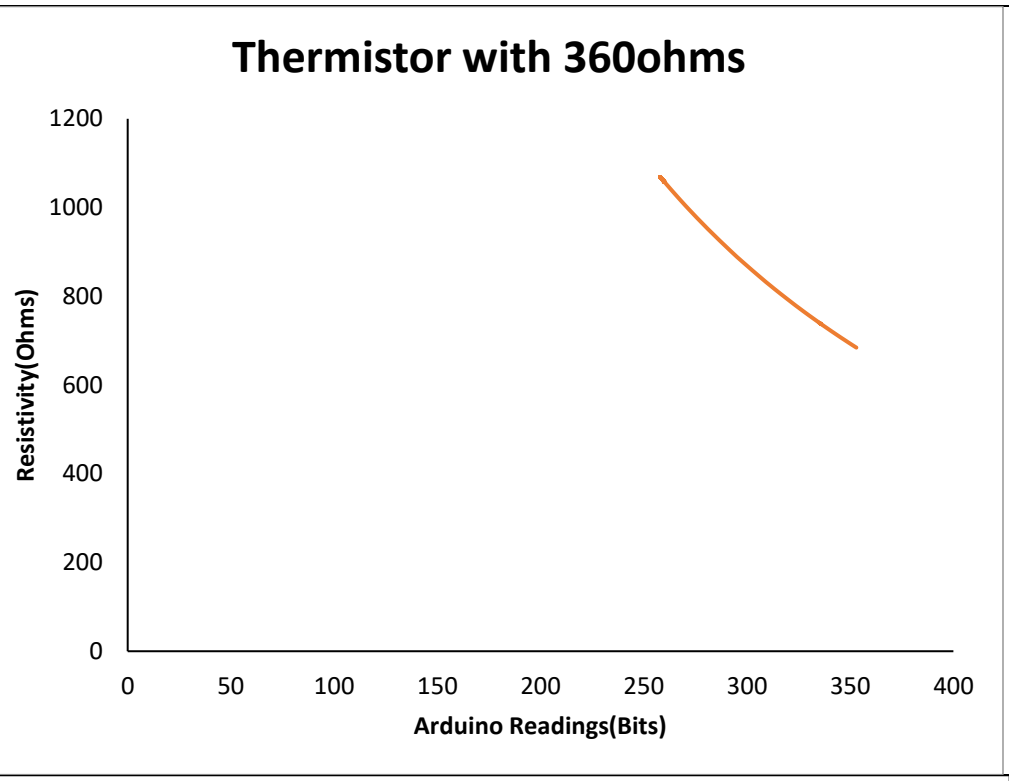
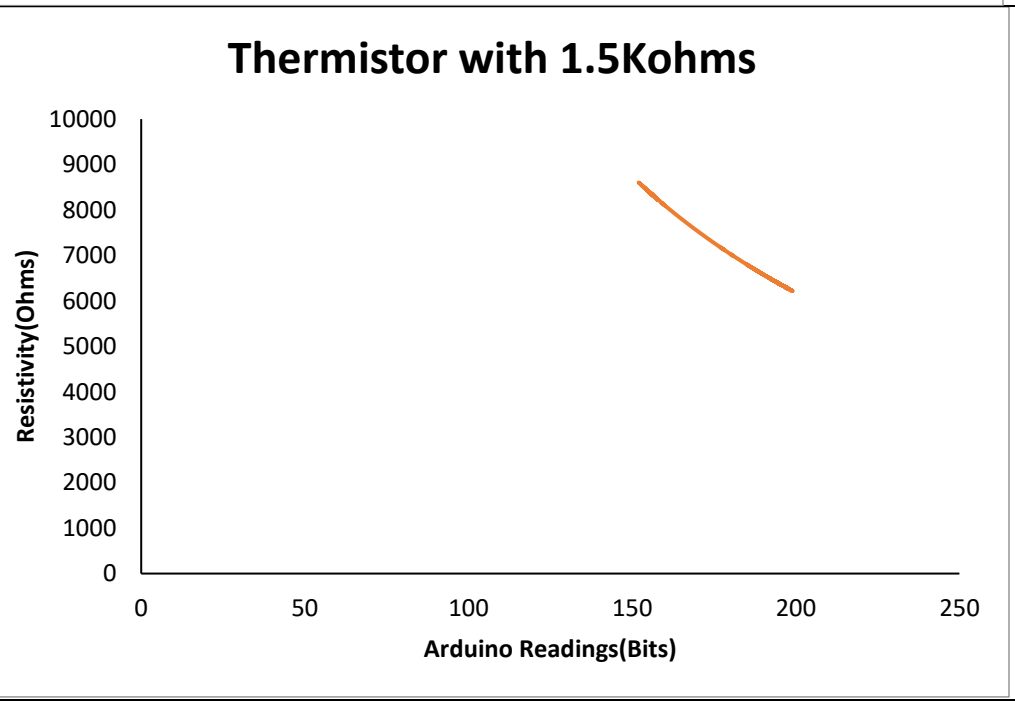
2. Resistor Selection for thermistor. (33%)
    a. Create a graph for each of the estimated resistances of the thermistor when pair with another passive resistor.
    b. Combine all the graphs and show the range of operation with a normalize voltage.
    c. Answer: Which passive resistor is better suited to be with the thermistor? What is the range of the resistor? What makes the resistivity to go down?

T120ohm	
T240ohm	

Lab 4 Report

Name: Tyler William Grindrod

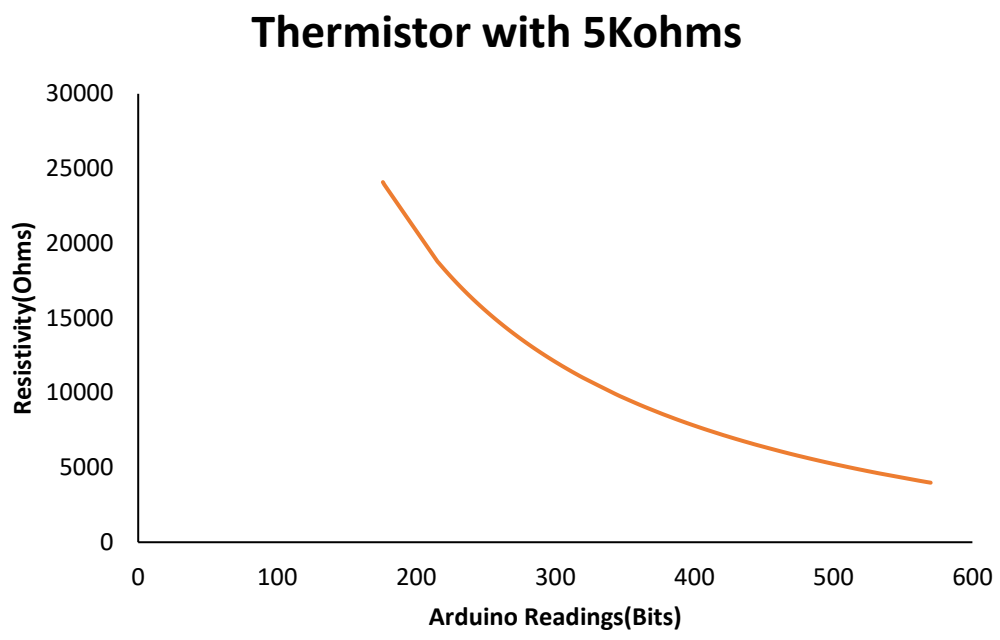
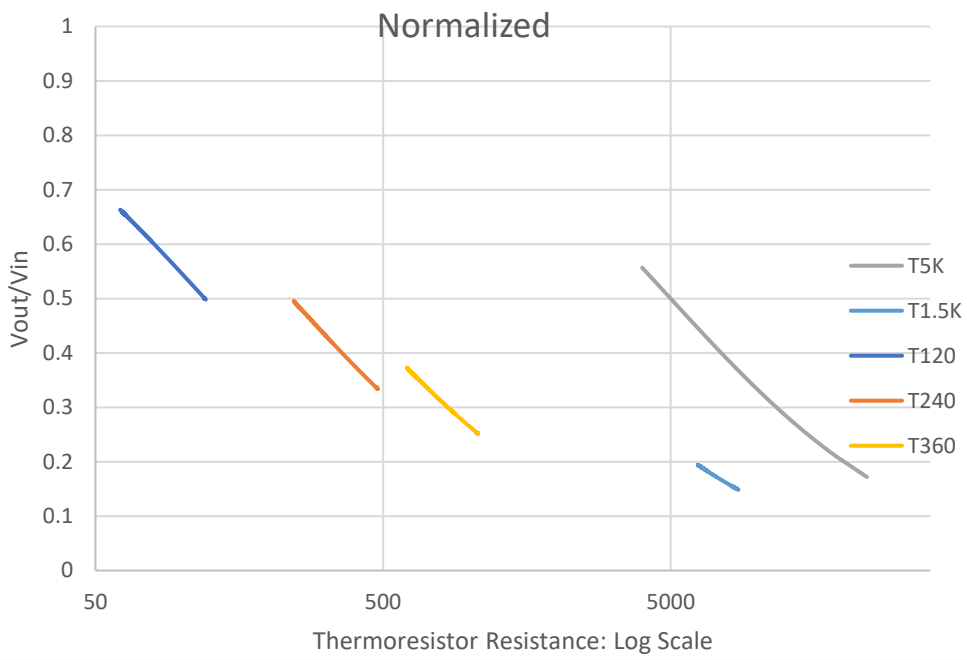
PID: 5695162 Date: 03/10/26

T360ohm	
T1.5Kohm	

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

T5Kohm	
Normalized (combined)	
Answer Part C:	In Voltage Divider the maximum change in Vout occurs when the Resistor that is fixed closely matches the resistace of our sensor. From the normalized graph above, There is a larger response (higher sensitivity) to temperature change from the **5k 0hm resistor**. So this our best suited fixed resistor. Its dynamic range is 394 ADC. The resistivity of the thermoresistor decreases as the sensor surface temperature increases, allowing more current to flow and increasing the output voltage in this configuration.
Code:	// Lab 5: ThermoResistor Sensor // Tyler W. Grindrod

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

	`#define ThermoResistor_Pin A0`
	
	`void setup() {`
	`  Serial.begin(9600);`
	`}`
	
	`void loop() {`
	
	`  int temperature = analogRead(ThermoResistor_Pin);`
	`  Serial.println(temperature);`
	`  delay(50); // wait 50ms between readings`
	
	`}`

Lab 4 Report

Name: Tyler William Grindrod

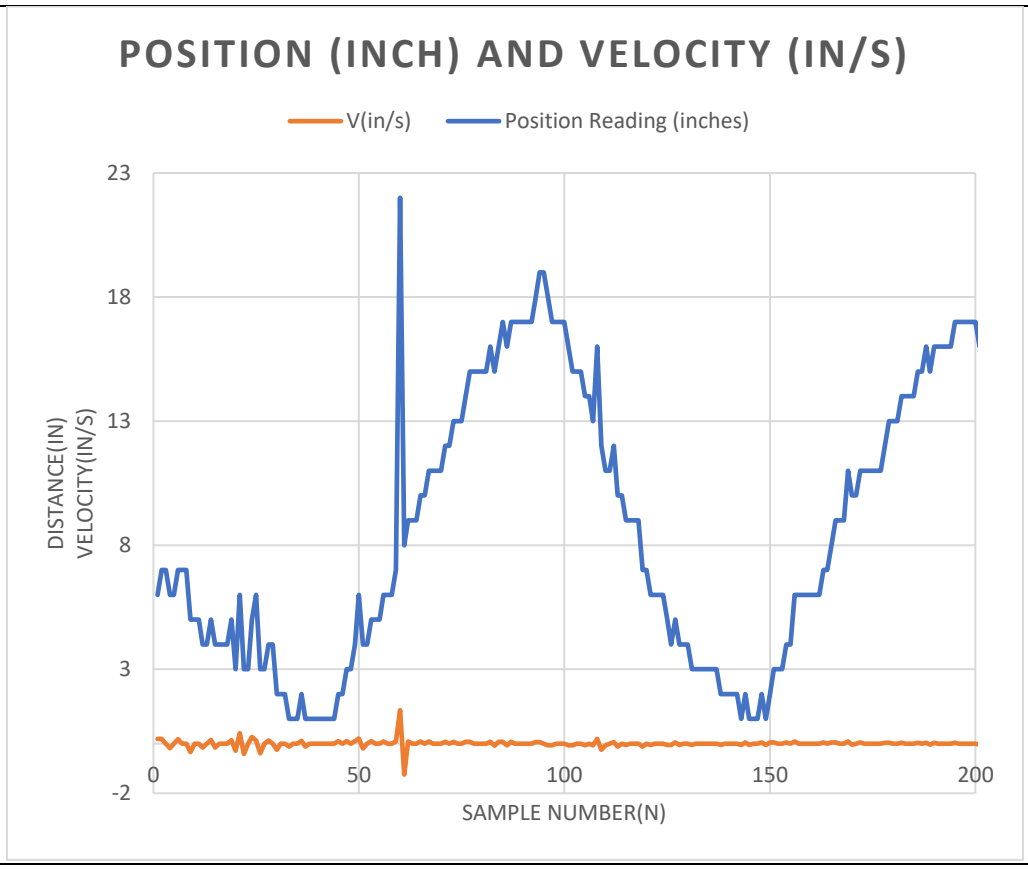
PID: 5695162 Date: 03/10/26

3. Distance and Velocity estimations. (33%)
    a. Report the position read by the Arduino in inches and in/s moving your hand slowly as a graph.
    b. Report the position read by the Arduino in feet and ft/s moving your hand slowly as a graph.
    c. Report the position read by the Arduino in inches and in/s moving your hand fast as a graph.
    d. Answer: Which of the measurements was better, in terms of usable data, feet or inches and why?

Lab 4 Report

Name: Tyler William Grindrod

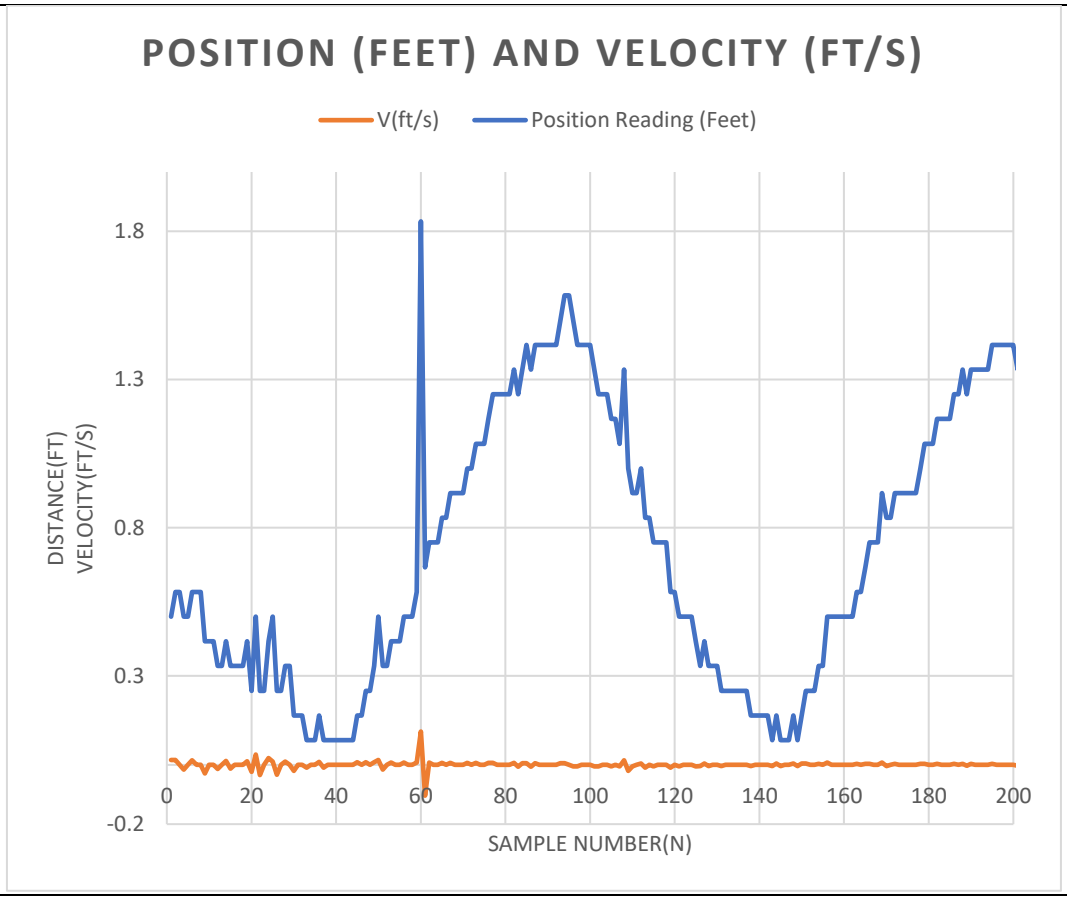
PID: 5695162 Date: 03/10/26

Slow Hand → in/sec	

Lab 4 Report

Name: Tyler William Grindrod

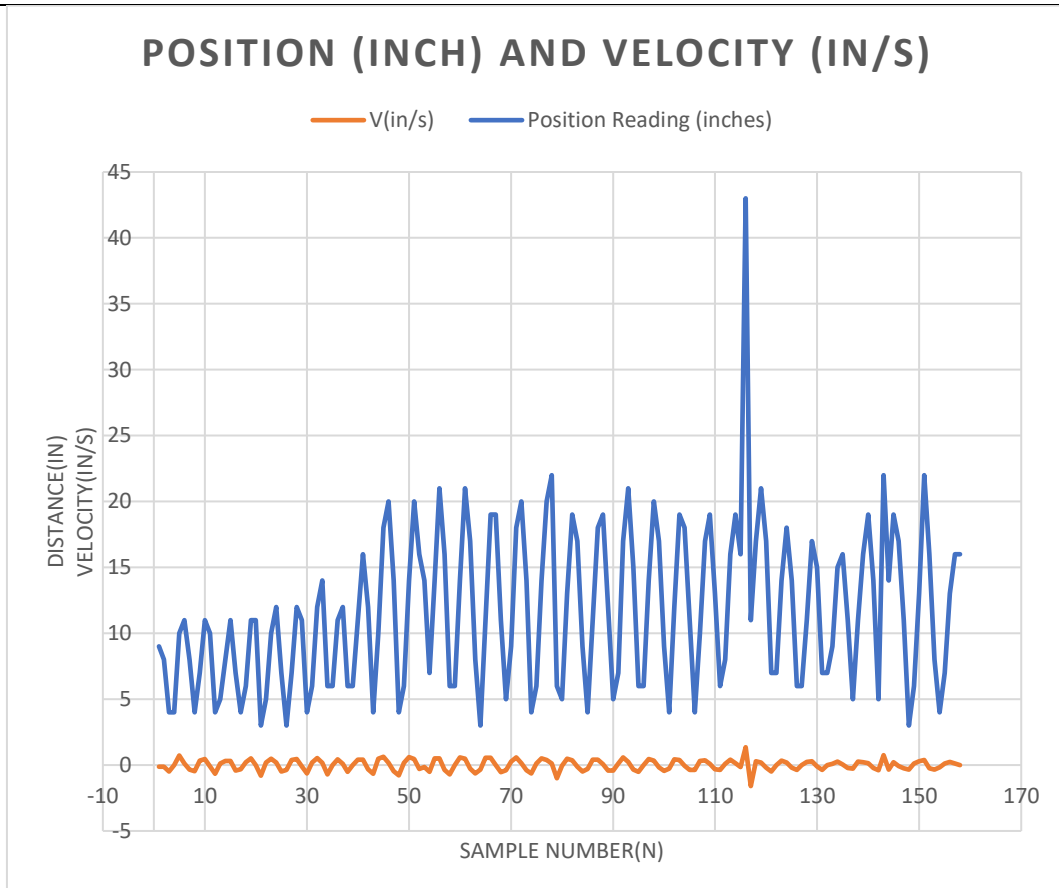
PID: 5695162 Date: 03/10/26

Slow Hand → ft/sec	

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

Fast Hand → in/sec	
Answer Part C:	**Inches** provided more useable data because the sensor's resolution is better suited for the smaller scale. In feet, the values are too small, leading to significant noise in the velocity calculations/ We obtained smoother results when measuring in in/s instead of ft/s.

Code:	// Lab 5: UltraSonic Sensor
	// Tyler W. Grindrod
	
	#define echoPin 8
	#define trigPin 12
	
	void setup() {
	  // initialize serial communication:
	  Serial.begin(9600);
	  pinMode(trigPin, OUTPUT);
	  pinMode(echoPin, INPUT);
	  Serial.println("Psotion_Inches, Time_ms");
	}
	
	void loop() {
	  long duration;
	  float posInches;

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

```
  unsigned long currentTime = millis();

  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);


  duration = pulseIn(echoPin, HIGH);
  posInches = (duration / 74) / 2;


  Serial.print(posInches);
  Serial.print(",");
  Serial.print(currentTime);
  Serial.println();

  delay(100);
}
```

Lab 4 Report

Name: Tyler William Grindrod

PID: 5695162 Date: 03/10/26

Objectives	Points	Full Credit	50%	25%	Earned
Electronic Diagram	34	1) Clear Graphs. 2) 1 graph per resistor and 1 normalized graph showing all the resistors used. 3) Proper Usage of resistors to find ideal. 4) Proper answering of the questions.	1) Resistors graphs. 2) Partial answered.	Resistors graphs.	
Assembly	33	1) Clear Graphs. 2) 1 graph per resistor and 1 normalized graph showing all the resistors used. 3) Proper Usage of resistors to find ideal. 4) Proper answering of the questions	1) Resistors graphs. 2) Partial answered.	Resistors graphs.	
Coding	33	1. Clear Graphs. 2. Cut off useless data. 3. Proper labeling of each scenario. 4. Proper answering of the questions	1. Graphs. 2. Proper labeling of each scenario. 3. Partial answers	Scenario graphs.	
				Total	